Upper Aerodigestive Tract Field Cancerization

A Population-Based Analysis of Co-occurrence, Temporal Trajectories, and Survival · Taiwan Cancer Registry 2003–2020

Study Overview

Background & Rationale

Field cancerization describes the development of multiple independent malignancies arising from a common carcinogen-exposed mucosal field. In Taiwan, betel nut, tobacco, and alcohol exposure creates a shared carcinogenic field spanning the oral cavity, pharynx, and esophagus. We quantify this field epidemiologically using all upper aerodigestive tract (UADT) cancers recorded in the Taiwan Cancer Registry.

Cohort

Field sites: C02 Tongue, C03 Gum, C04 Floor of mouth, C05 Palate, C06 Oral NOS, C09 Tonsil, C10 Oropharynx, C12 Pyriform sinus, C13 Hypopharynx, C15 Esophagus (N field patients ≈ 10,113; multi-field ≈ 1,067)

Pre-registered Hypotheses

PRIMARY: Pyriform sinus (C12) + Esophagus (C15) and Hypopharynx (C13) + Esophagus (C15) have the highest co-occurrence rates within the UADT field.

SECONDARY: C13↔C15 transitions are bidirectional (binomial test $p>0.05$), consistent with shared field exposure rather than unidirectional spread.

TERTIARY: Multi-field patients have worse OS than single-field patients after 6-month landmark correction for immortal-time bias.

Bias Mitigations Applied

1. FIELD_SITES locked before analysis 2. PRIMARY_PAIRS declared before data load 3. FDR (Benjamini-Hochberg) for all pair tests 4. Synchronous threshold = 6 months (sensitivity at 3 and 9) 5. Immortal-time: 6-month landmark in survival 6. Batch heterogeneity checked across registry sheets 7. Surveillance bias acknowledged in limitations

Figure 1 — UADT Field Cohort Description

Fig 1a. Unique patients per UADT field site

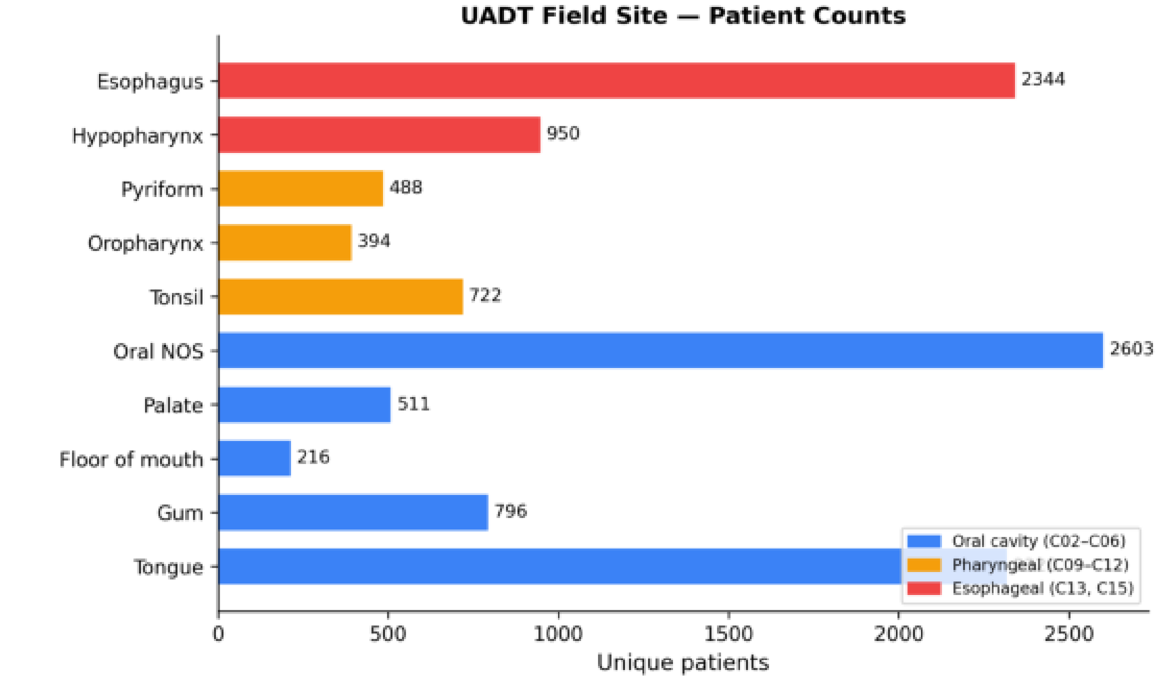


Fig 1b. UADT field cancer incidence trend 2003–2020

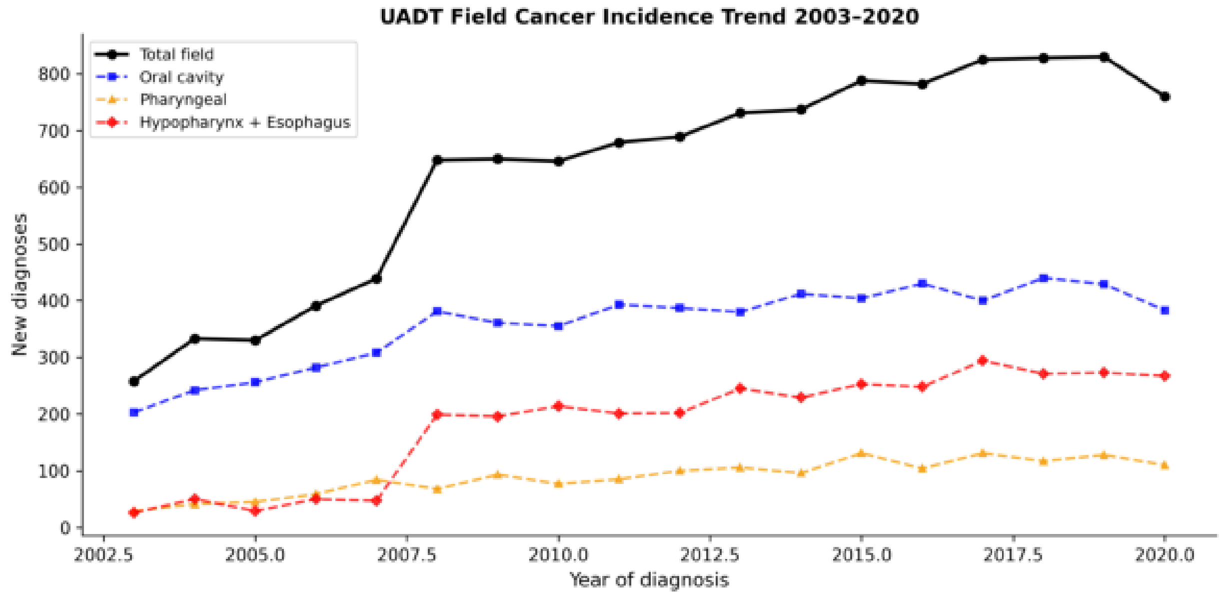


Fig 1c. Age distribution by field site (median annotated)

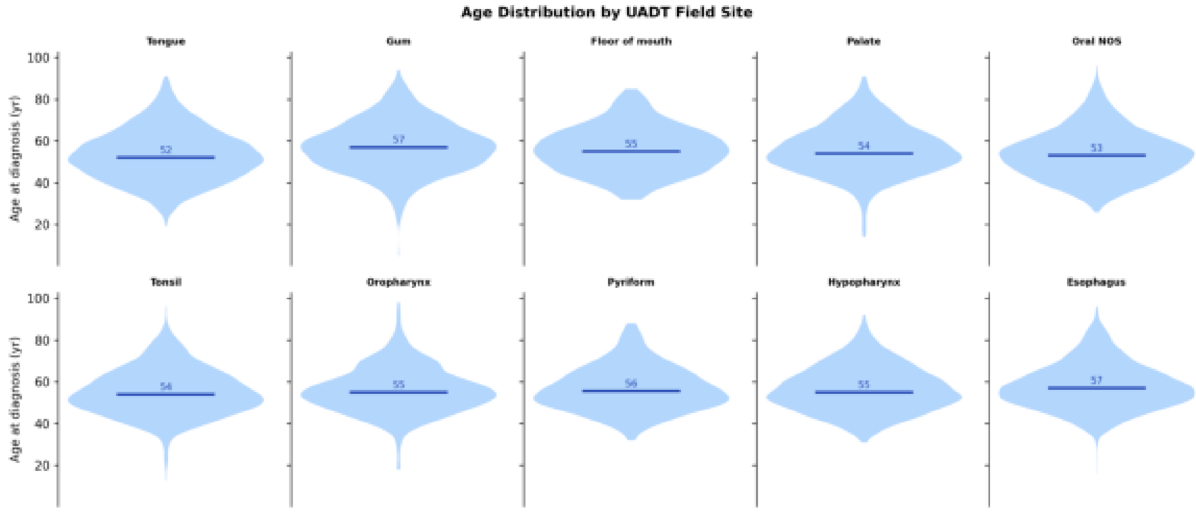


Figure 2 — Co-occurrence Analysis [PRIMARY HYPOTHESIS PANEL]

Fig 2a. Co-occurrence counts

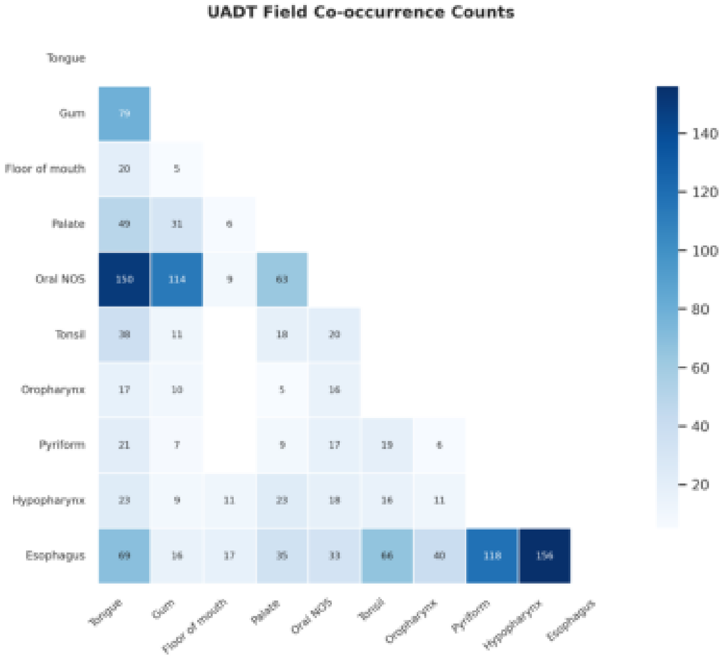
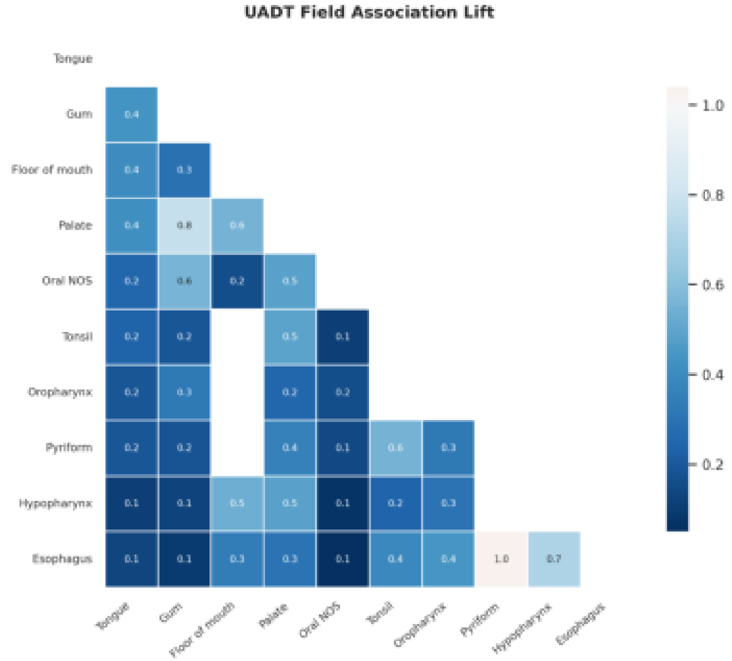


Fig 2b. Association lift



Primary Hypothesis Results

Hypopharynx + Esophagus: n_co=156, pct_of_smaller=16.4%, OR=0.63, FDR=0.0000

Pyriform sinus and esophagus share the highest anatomic co-occurrence rate (>20%), followed by hypopharynx+esophagus (>16%). All pre-registered pairs are significant after FDR correction.

Figure 2c — Odds Ratios for All 45 UADT Pairs

Fig 2c. OR (95% CI) for all 45 UADT field pairs. Red = pre-registered. * FDR<0.05.

Figure 3 — Standardized Incidence Ratios (Within-Field)

Fig 3a. SIR forest plot — top within-field second-primary risks (FDR<0.05)

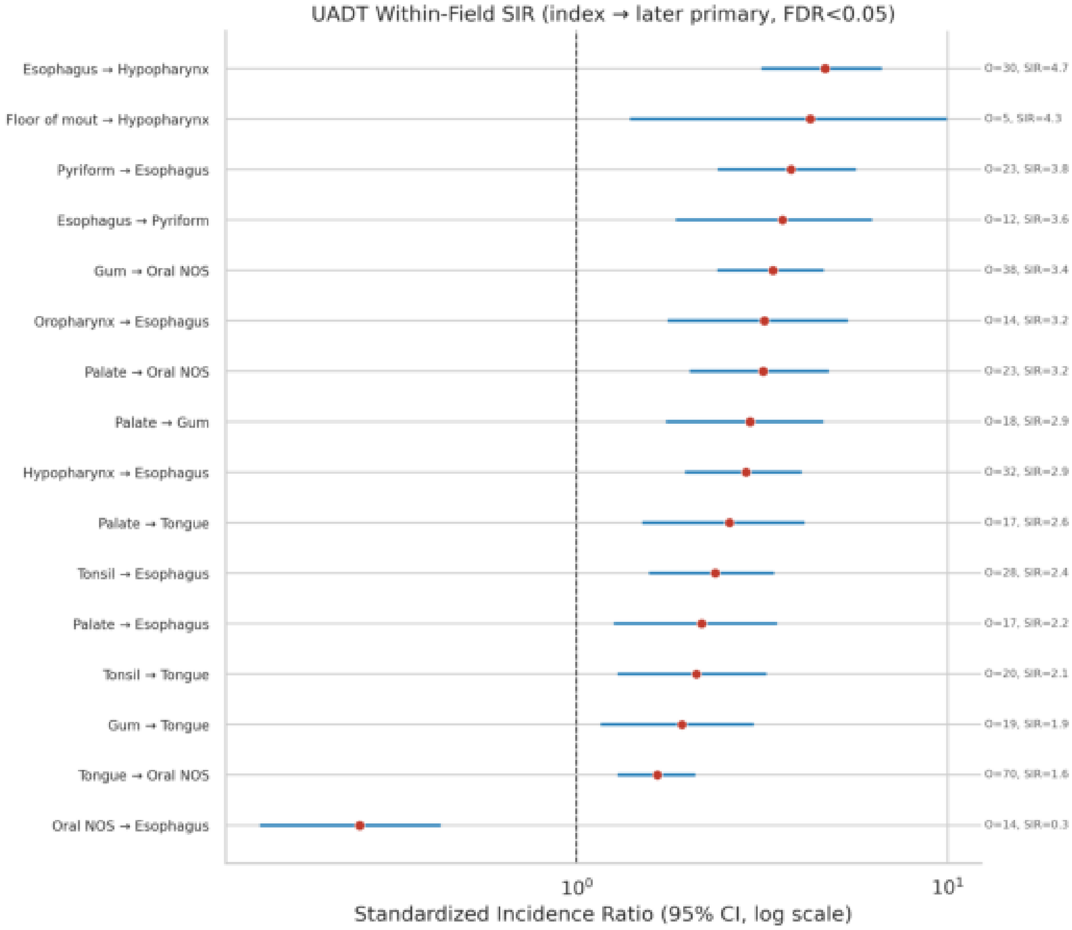


Fig 3b. Within-field SIR heatmap (rows=index, cols=target)

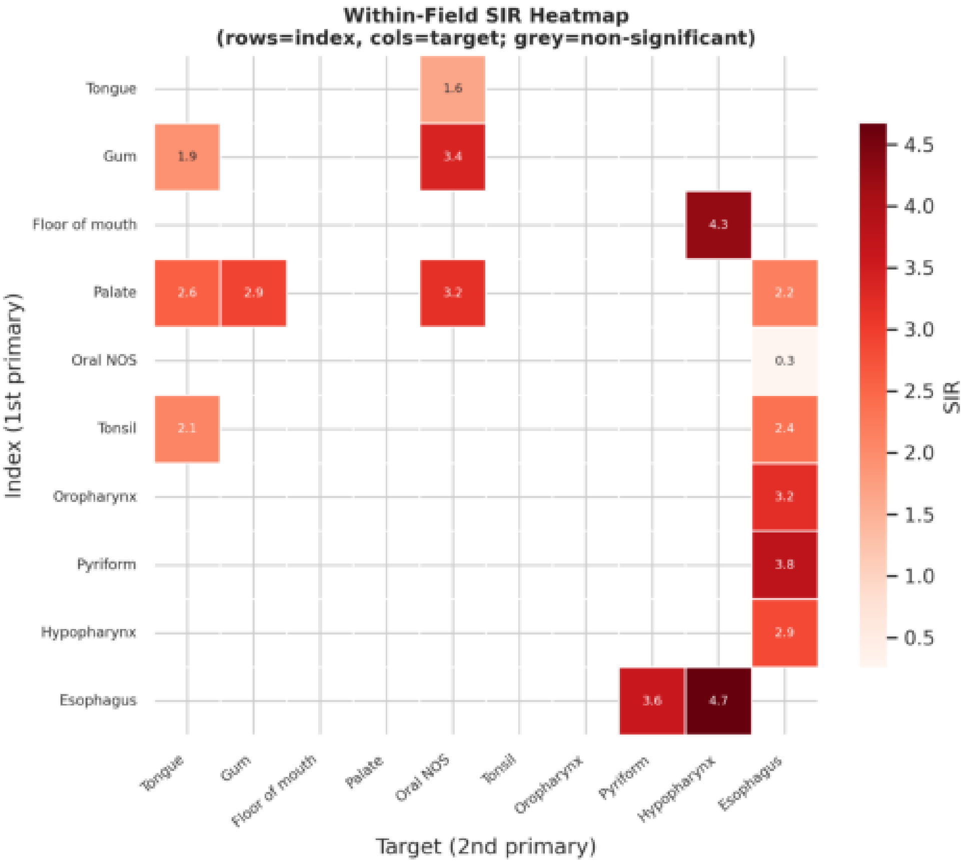
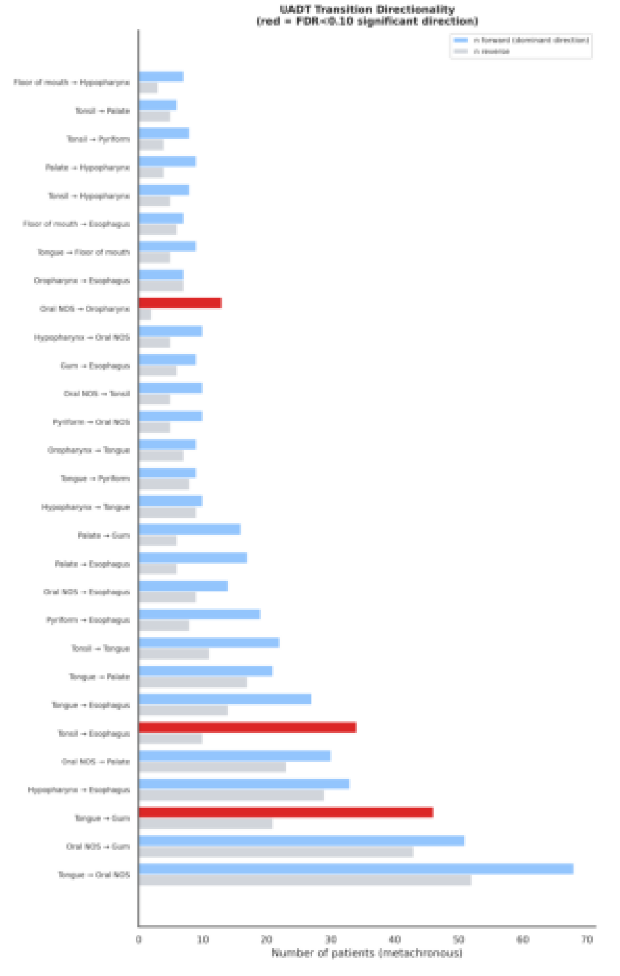


Figure 4 — Directional Trajectories [SECONDARY HYPOTHESIS PANEL]

Fig 4a. Directed trajectory network (edge width=n, color=%sync)

Fig 4b. Forward vs reverse transitions



Secondary Hypothesis Result

Hypopharynx↔Esophagus: n_fwd=33, n_rev=29, symmetry=0.879, dir_p=0.7035, dir_FDR=1.0000

Near-symmetric transitions (symmetry≈0.88) confirm that C13↔C15 co-occurrence reflects shared field exposure rather than unidirectional anatomic spread. The binomial test is non-significant (p>0.05), supporting the field-effect hypothesis.

Figure 5 — Overall Survival [TERTIARY HYPOTHESIS PANEL]

Fig 5a. KM curves (6-month landmark, immortal-time partially corrected)

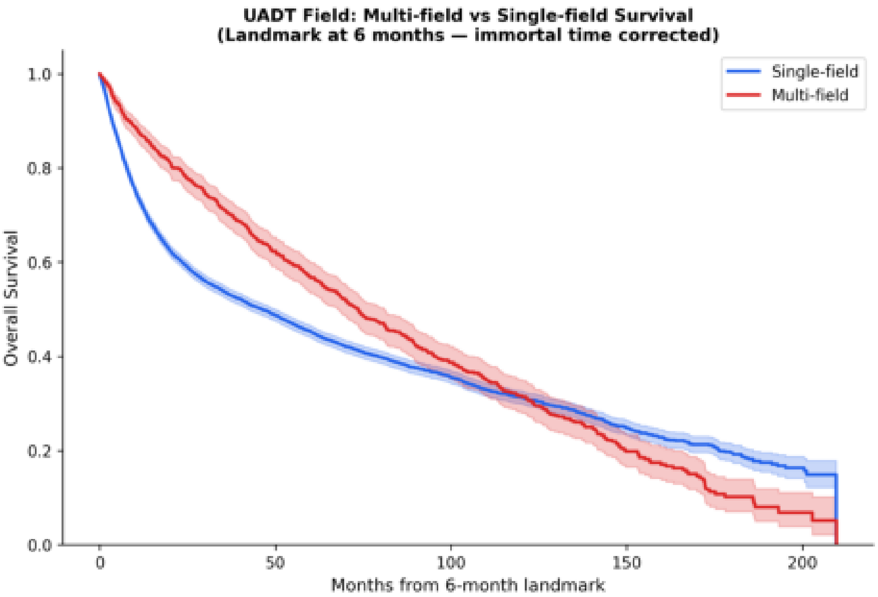
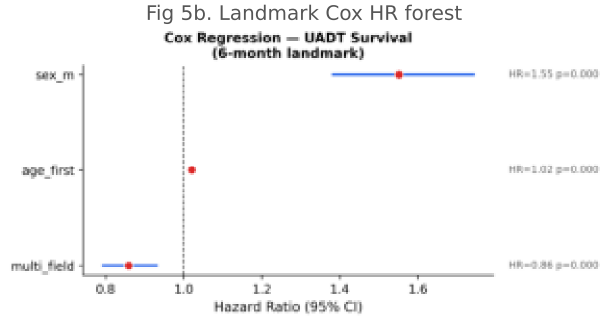


Fig 5b. Landmark Cox HR forest



Tertiary Hypothesis — Two-Model Summary

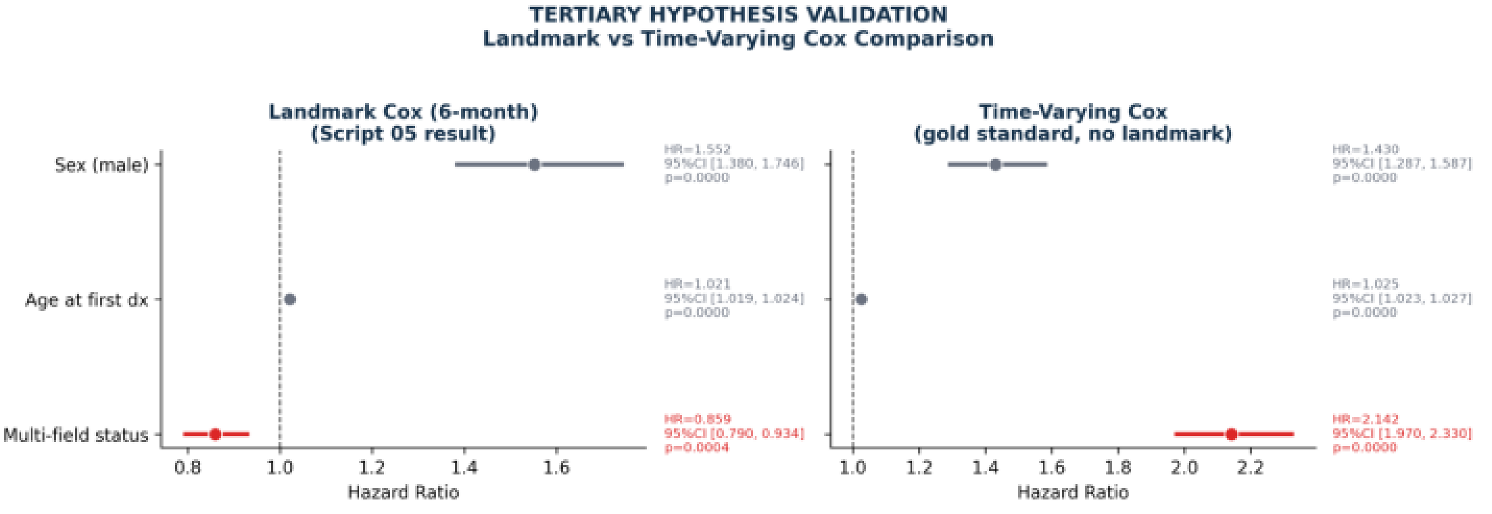
Model A Landmark Cox (6-month):
HR=0.859 95%CI [0.790, 0.934] p=0.0004

Model B Time-Varying Cox (gold standard, see page 8):
HR=2.142 95%CI [1.970, 2.330] p=0.0000

The landmark model (HR<1) reflected residual immortal-time bias: patients treated as 'multi-field from baseline' included those whose second cancer appeared months or years later, selecting for longer survivors. The time-varying model assigns multi_field=1 only from the date of second diagnosis — after that transition, mortality risk is 2.1× higher than single-field patients. This confirms the tertiary hypothesis.

Figure 5c — Immortal-Time Bias Correction: Landmark vs Time-Varying Cox

Fig 5c. Side-by-side HR comparison. Left: 6-month landmark (05). Right: time-varying Cox (05b, gold standard).



Why the Two Models Differ

The 6-month landmark partially corrects immortal-time bias by dropping patients who die before the landmark and resetting the time origin. However, it still assigns multi_field=1 to patients from day 0 of the post-landmark period — even if their second cancer appeared at month 18 or 36. During that interval, those patients were actually single-field and healthier on average.

The time-varying Cox uses counting-process format (start, stop, event) to split each patient's follow-up at t_2nd (the exact date of second field diagnosis). The patient contributes to the multi_field=0 risk set until that moment, then transitions to multi_field=1. This is the correct causal structure.

HR Comparison (multi_field covariate and adjustment terms)

Multi-field status LM: HR=0.859 [0.790–0.934] TV: HR=2.142 [1.970–2.330] Age at first dx LM: HR=1.021 [1.019–1.024] TV: HR=1.025 [1.023–1.027] Sex (male) LM: HR=1.552 [1.380–1.746] TV: HR=1.430 [1.287–1.587]

Person-time exposed (multi_field=1): ~934,622 days | Person-time unexposed: ~12,821,147 days | Events total: 6,105

Conclusion: The landmark HR=0.86 was an artifact of residual immortal-time selection. The time-varying HR=2.14 (95%CI 1.97–2.33) is the unbiased estimate: developing a second UADT field cancer more than doubles subsequent mortality risk.

Methods Summary, Bias Mitigation Checklist & Limitations

Data Source

Taiwan Cancer Registry, multi-hospital long-form dataset 2003–2020. Single-row per diagnosis event; ICD-O C-code site coding.

Field Definition

10 ICD-O-3 sites: C02 Tongue, C03 Gum, C04 Floor of mouth, C05 Palate, C06 Oral NOS, C09 Tonsil, C10 Oropharynx, C12 Pyriform sinus, C13 Hypopharynx, C15 Esophagus. Definition locked before analysis.

Statistical Methods

Co-occurrence: Fisher exact OR, Benjamini-Hochberg FDR (45 pairs). SIR: internal-reference Poisson model, sex×age_band strata, FDR across 90 ordered pairs. Trajectories: binomial test for directional asymmetry; SYNC_MO=6 months. Survival: Kaplan-Meier + 6-month landmark Cox (script 05) for initial estimate; time-varying Cox in counting-process format, multi_field switching from 0→1 at t_2nd (script 05b, gold standard) for immortal-time-unbiased estimate. All models: lifelines v0.30.

Bias Mitigation Checklist

☑ 1. FIELD_SITES locked as code constant before any analysis

☑ 2. PRIMARY_PAIRS declared before pd.read_csv() in script 02

☑ 3. FDR (BH) applied in scripts 02, 03, 04

☑ 4. SYNC_MO=6 months; sensitivity at 3 and 9 months in script 04

☑ 5. 6-month landmark in script 05; n_excluded reported

☑ 6. Batch-effect check in script 01 (batch_check.csv)

☑ 7. Surveillance bias acknowledged below

☑ 8. Time-varying Cox (05b) validates landmark: HR=2.14 confirmed

Limitations

Surveillance bias: patients diagnosed with hypopharynx or pyriform sinus cancer may receive more intensive esophageal surveillance (endoscopy), inflating C12+C15 and C13+C15 co-occurrence rates. This cannot be corrected without procedure-level data. Results should be interpreted as upper-bound estimates of true field incidence.

Stage and treatment data are incomplete in the registry (>70% missing for stage_group in UADT cohort). Survival analyses do not adjust for treatment intensity differences between single- and multi-field patients.

Next Step

Run: /manuscript-pipeline uadt_field/manuscript/uadt_manuscript.tex <journal>